CODING FOR DATA SCIENCE AND ANALYTICS

Exercise Session 3: Randomness, Loops and Sentinels

Exercise 1 — Descriptive Statistics

Given the following list of daily temperatures:

```
temperatures = [22.5, 24.1, 19.8, 25.3, 21.0, 18.5, 23.2]
```

Goal

Compute the following descriptive statistics: **minimum**, **maximum**, **interval**, **count**, **sum**, **mean**, and **variance**.

Constraint

Do not use Python's built-in functions (e.g., `min()`, `max()`, `sum()`, `len()`).
You can iterate through the list using a simple `for element in list:` loop.

Exercise 2 — Smart Home Automation

Write a program that controls a Smart Home's lights and blinds based on the current weather or a special mode. The system should continuously ask the user for input until they type the sentinel word "shutdown".

Rules:

- **sunny** → close blinds, turn on *cool white* light.
- **cloudy** → open blinds, turn on *warm* light.
- **rainy** → open blinds, turn on *cozy yellow* light.
- **movie** → ignore the weather: close blinds, turn on *purple* light.

Hint

Use a `while` loop controlled by a Sentinel variable to keep the system running.

Exercise 3 — Stock Market "Random Walk"

Let's simulate a financial investment using a random walk.

- **Initial price:** $100.
- **Stop-loss:** $80.
- **Take-profit:** $130.

Goal

Each "day", the stock price changes by a random percentage between -5% and +5%. Simulate the days passing until the price hits or crosses either the stop-loss or the take-profit.

Requirements & Bonus

Print the final outcome (Profit or Loss), the final price, and how many days the position remained open. Use a sentinel variable to control the loop.

Bonus: Try to create a simple ASCII chart using `print()` to visualize the daily price trend! (Hint: use string multiplication like `" " * n`).

Exercise 4 — The Lucky Game

Implement a dice game following these rules:

- Roll two 6-sided dice.
- First roll: If the sum is 7 or 11, you WIN. If it is 2, 3, or 12, you LOSE.
- Otherwise, the sum becomes your *target*. You must keep rolling the dice until you roll the target again (WIN) or roll a 7 (LOSE).

Hint

Use the `random` module to simulate the dice rolls. Setting a seed (`random.seed(42)`) can be useful for reproducibility during your testing.